EXPERIMENT 6

load_balancing.py Python

```python
# Threshold t1 and t2 Load Balancing Algorithm

# Example initial server loads
servers = [2, 4, 6, 3]

# Taking thresholds as input
threshold1 = int(input("Enter Threshold t1 (Normal limit): "))
threshold2 = int(input("Enter Threshold t2 (Critical limit): "))

def assign_request_to_server(request_id):

    for i in range(len(servers)):

        # Case 1: Server under normal load
        if servers[i] < threshold1:
            servers[i] += 1
            print(f"Request {request_id} assigned to Server {i}")
            print(f"Current Load: {servers[i]} (Below Threshold1)\n")
            return

        # Case 2: Server busy but still acceptable
        elif threshold1 <= servers[i] < threshold2:
            servers[i] += 1
            print(f"Server {i} is busy but accepting request {request_id}")
            print(f"Current Load: {servers[i]} (Between Threshold1 and Threshold2)\n")
            return

        # Case 3: Server overloaded
        else:
            print(f"Server {i} overloaded (Load = {servers[i]})")

    print("All servers overloaded. Request rejected.\n")


print("Initial Server Loads:", servers)
print()

# Simulate requests
for req in range(1, 6):
    assign_request_to_server(req)

print("Final Server Loads:", servers)

```

Output:

```
PROBLEMS    DEBUG CONSOLE    TERMINAL    PORTS    OUTPUT                                zsh - expt_6

➜  expt_6 git:(main) ✗ python3 load_balancing.py
Enter Threshold t1 (Normal limit): 4
Enter Threshold t2 (Critical limit): 6
Initial Server Loads: [2, 4, 6, 3]

Request 1 assigned to Server 0
Current Load: 3 (Below Threshold1)

Request 2 assigned to Server 0
Current Load: 4 (Below Threshold1)

Server 0 is busy but accepting request 3
Current Load: 5 (Between Threshold1 and Threshold2)

Server 0 is busy but accepting request 4
Current Load: 6 (Between Threshold1 and Threshold2)

Server 0 overloaded (Load = 6)
Server 1 is busy but accepting request 5
Current Load: 5 (Between Threshold1 and Threshold2)

Final Server Loads: [6, 5, 6, 3]
➜  expt_6 git:(main) ✗
```